

Unit 1 : Introduction to Product Design and Development

Definition of product design, Essential factors for product design, Modern approaches to product design, Standardization, simplification and specialization in product design, Product development versus product design ,Modern product development process, Product testing and validation.

Product Design

- A design process is a set of technical activities within the product development process that works to meet the marketing and business case vision.
- Product design is also defined as to create a new product to be sold by business to its customer. It is a set of strategic and technical activities from idea generation to commercialization.
- Product design includes all the engineering and industrial design work that goes into developing a product from the initial concept to production, which ensures that it works reliability, is cost effective for manufacturing and it should look good.
- The role of design is to create a successful marketable product from an innovation. The uniqueness of design is often the deciding factor in the success of product.

Types of design

There are generally 6 types of engineering design:

Selection design

In selection design, the design engineer simply picks out an existing solution that meets the specifications and requirements of the design problem. An example is choosing a proper bearing for a wheel. The key to a successful selection design is gaining a full understanding of both the design need as well as the limitations of existing solutions.

Configuration design

When a computer company puts together a made-to-order computer, they are doing configuration design. Standard components are selected and combine to make a new configuration to meet unique requirements that old configurations cannot meet.

Parametric design

In parametric design, an optimal design solution is found by finding proper parameter values. A simple example is the design of a rope bridge the rope size is a parameter that is varied until the bridge can support the necessary weight.

Routine design

Routine design may also be referred to as "cookbook" designs. There are many standards and codes that outline the steps to follow and calculations to perform for the design of many different items. A common example is using the Boiler and Pressure Vessel Code (American Society of Mechanical Engineers) to design large water tanks.

Redesign

In redesign an existing solution already exists and a better solution is sought. The better solution may be driven by changing needs, the desire to reduce costs, or the availability of new technology. A classic school example of redesign is to "design a better mousetrap".

Original design

Original design begins with a stated need and only a completely blank sheet of paper. This type of design arises from a completely new idea that has no apparent design solution.

Product Development

- Product development is a set of activities required to bring a new concept to a state of market readiness.
- This is also called as new product management, product development is the conceptualization, design and development and marketing of newly created or newly rebranded goods.
- This set includes everything from new product vision to business case analysis activities, marketing efforts, engineering design activities, development of manufacturing plans and the validation of product design to confront to these plans. Often it also includes development of channels for marketing and introducing the new product.
- Product design set includes refinement of product vision into technical specification, new concept development and embodiment engineering of new product.
- Product design does not include all of business and financial, management activities of product development.
- The design of a manufacturing process is considered as part of product development.

Product Development Team Functions

The Belbin team role model is designed to use the talents and personalities of team members. If you want your team to improve its performance by working together more effectively, it is important that you learn to use the psychological diversity in your team to its fullest potential. Belbin has defined Nine Roles for team as follows,

1. Implementer is the practical organizer within the team. Disciplined, orderly and task-oriented. Implementer puts plans and ideas into easily executable tasks. He/she can be a little too practical and conservative when the usefulness of new ideas is not immediately clear.
2. Resource Investigator is the cheerful, extroverted person with lots of contacts within and outside the team. He/she is enthusiastic, adventurous and open-minded and always looking for new ideas. Resource Investigator is naturally good at developing and maintaining contacts, but may become careless when the novelty wears off.
3. Plant is the creative thinker of the team. Innovative and original. A free spirit who needs space to fantasize about new and surprising solutions to complex problems. Plant is not always practical and sometimes misses what others require from him/her.
4. Monitor is sensible, thoughtful and critical. The analyst of the team. He/she is always analyzing situations and wants to get to the bottom of things. As Monitor he/she may deliberate long and thoroughly and his or her judgment is rarely wrong.
5. Shaper is driven, passionate and willful. He/she has a strong urge to perform, looks for challenges and gets things going. Shaper makes sure deadlines are made and goals are met, one way or the other. He/she can get frustrated and react angrily or emotionally.
6. Coordinator is the natural coordinator of the team. He/she looks after procedures, helps team members clarify intentions and summarizes what everyone wants. He/she has a

nose for talent and knows how to utilize people to their full potential. Coordinator trusts others and delegates easily but sometimes has a tendency to leave too much work to others.

7. Completer Finisher has the talent to always feel what could go wrong. This leads to a lot of attention to details, checking and rechecking, and a tendency to perfectionism. He/she monitors the quality and safety, but can sometimes be overprotective and find it difficult to delegate things to others.
8. Team worker is the most sensitive member of the team, helpful, accommodating and focused on creating a pleasant atmosphere and sense of togetherness. He/she prefers balance and harmony and is close to others. Team worker has difficulty with conflict and may have trouble taking decisions at critical times.
9. Specialist is an immeasurable source of knowledge in his/her field. An advisor who is gladly consulted and will provide knowledge on a specific topic with pleasure and ease. His/her contributions to the team are his/her substantial knowledge and technical skills. Specialist thrives less well when working together and does not care for social activities that much.

Essential Factors for Product Design

There is a large number of factors which influences product design. These factors can be broadly classified in four groups.

- i) Technical factors.
- ii) Industrial design factors.
- iii) Designing for production-economic factor.

i) Technical Factors:

a) Operating condition: The product designer should consider the type of person (men, women, infant, children, skilled, unskilled and the like) who would use the product and the environmental conditions in which the product will be used. E.g. The noise level. vibration level at which the product will be operated should be given due consideration, while designing product.

b) Performance parameters: This consider speed feed etc. at which the product will be used. The accuracy and quality in performance needed and expected duration of continuous satisfactory working of product. A product is designed to give certain level of performance.

c) Maintenance factors: The product design will be affected by maintenance factors like how often maintenance and repair can be undertaken, whether planned or breakdown maintenance will be adopted by the user etc. Product design vary with respect to the way, the product can be maintained.

ii) Industrial Design Factors:

a) Functional aspect : Product design vary with regards to functional scope of the product. Following questions needs to be answered for considering functional aspects, which affects the product design. The answers to these questions, will vary from person to person and time to time. A designer with creative mind will invariably come up with new or modified product designs.

b) Operational aspects: After considering the functional aspects, the operational aspects should be considered. A product design should be such that, the final product would be easy to handle and simple to operate. It should be adopted to various operational conditions. Product should be designed for increased versatilities.

c) Aesthetic aspect: This concerns the final shape appearance and the size of the product. Aesthetic makes the product attractive and include in the customer willingness to possess or purchase. The appearance of product makes design impressions on a buyer. If these are two products having approximately the same function, the same cost and the same time required for delivery, the product with better appearance will have a greater sales appeal. e.g. contrast, balance, surface treatment (colour and surface texture).

d) Human factors: In order to have an effective human-compatible product, the relevant human factors must be carefully be considered during the design stage.

- The main objective should be to design a product that allows people to use it in most suitable manner, that possess adaptability to humans and it should be ensured that product does not subject the user with extreme physical or mental stress or to any hazard.
- Present day automobile design process allows for greater visibility, easy to read displays, more leg space, seat adjustment for driver and other occupants comfort.

Planning Tools

Following are planning tools

1. Affinity Diagram (KJ method)
2. Interrelationship diagram
3. Tree diagram
4. Prioritization matrix
5. Matrix diagram or quality table

List of Scheduling Tools

Following are scheduling tools

1. GANTT Chart
2. Schedule Network Analysis
3. Critical Path Method
4. PERT (Program Evaluation and Review Technique)
5. Schedule Compression

Modern Approaches to Product Design

- Now a days, product design because more important than ever because customers are demanding greater product variety and are switching more quickly to the products with the state of the art technology.
- Organisations are adopting modern approaches to the product design such as mass customization design for manufacturing and assembly, product disposal, quality function deployment and time based competition.

Mass Customization:

- Mass customization is the low cost, high quality, large volume delivery of individually customized products. It is ability to quickly design and produce customized products on large scale at a cost comparable to non customized product.
- Customer is expected to receive customized products at the cost close to mass production cost. As market become more and more segmented and aggregate demand remain constant or increases. Mass customization design approach will focus on to reduce the time required to deliver customized product and to reorganise design and production process quickly in response to customer requests.
- It would be counter productive to opt for mass customization, if a customized process takes too long time to produce speed is very important factor in design and production process which is a criterion for evaluation of organization's mass customization capability.

Design for Manufacturing and Assembly (DFMA)

- Manufacturing of a newly designed product is an important task of product design process. Design for Manufacturing and Assembly (DFMA) can be powerful tool to improve product quality, manufacturing cost and time. This approach focuses on manufacturing issues during product design.
- DFMA is implemented with the help of computerized softwares that recognises design concepts considering economic implementations of design decisions. These decision are very important part because product design decisions fix 70 to 90% of manufacturing cost.
- To do this effectively, firm must concentrate on earlier design stage about service and maintenance issues.

Disposal Approach

- While designing a product, because of government rules and regulations becoming an important part of product design. This requirement of disposing manufactured product has become major requirement of product design. This approach changes the notion that consumer is the final owner for a product.
- With this approach product return to manufacturer for recycling and the recycling process should begin in product design. Product should be designed in such a way that they can be disassembled and recycled easily.
- The design engineers should avoid exotic materials that are difficult to recycle. For example parts with plastic and metal fused together should not be used, where they are difficult to separate. The designer should determine, which parts should be designed to be reused and which will be designed to be discarded or broken down and recycled. All these should be done without adding any extra cost or reducing product quality.

Quality Function Deployment:

- Product design dictates the product's quality. Quality defines the way that goods and service functions. Quality has at least two essential components.
- First the product must be designed in such a way that it should function with a high probability of success or reliability. It should perform specific function without failure under given conditions. When product reliability increases, manufacturer can extend the products warrantee without increasing customer claims for repairs or returns. Warrantee for expensive items such as mobile phones or electronic appliances are important selling points for customer.
- Second, quality improves when operating or performance characteristics improve even though reliability does not. Quality function deployment is being used by organisation

to translate customer wants into working products. This is also referred as house of quality.

Principles of Product Design and Development

The standardization, simplification and specialization are the three basic principles of product design and development.

i) Standardization:

- Product standardization is a principle in product design and development that focuses on to reduce the number different parts within a product.
- Standardization is the base of all mass production. They make possible thousands of different parts to be placed within the rich of energy body. E.g.: Spark plug of a two wheeler, if purchased of any make or manufacturers will fit exactly into engine head.
- Standardization means producing maximum variety of products from the minimum variety (e. standardized) materials, parts, tools and processes. This reduces the cost for manufacturing and make product economical. Standardization means that non-standard products will not be produced except when customer orders them to be made.
- Standardization is the process of establishing standards or units of measure by which extent, quality, quantity, value, performance etc, may be compared and measured.

Advantages:

- i) Reduces the work to design the specifications, drawings and part details of energy component.
- ii) The time saved can be utilized for developing new or to improve previously designed product.
- iii) Lesser design mistakes and design alterations.
- iv) Lower unit cost.
- v) Better quality products.
- vi) Accurate delivery dates.
- vii) Increased interchangeability of parts.
- viii) Better utilization of manpower and equipment.

Disadvantages:

- i) Reduction in choice because of reduced variety and consequent loss of business.
- ii) Change in public taste seriously affects a company producing only standardized product range.
- iii) It becomes difficult to introduce new models because of less flexible production facilities.
- iv) Standards once set resist change and thus standardization may become an obstacle to progress.

Applications:

1. Finished products e.g. Cars, televisions.
2. Sub assemblies and components: Automobile clutches, brakes or lights.

3. Material standardization, eg. both of direct materials (plain carbon and alloy steels, are welding electrodes etc.) and indirect materials (such as oil and greases).
4. Production equipment standardization. e.g. that of machine tools, presses, welding equipment's etc.

Simplification:

- The concept of simplification is closely related to the standardization. This is the technique of reducing variety of products manufactured. This aims to reduce product range, assemblies, parts materials and design.
- A manufacturer may reduce the number of different types of two wheelers from a dozen to three or four to simplify his range.
- Simplification removes the superfluous. It decreases variety of sizes, for example a bottle making company making plastic boules of sure 100 ml, 150, 200, 250, 350, 450, 500, 600, 750, 1000 ml can eliminate superfluous stress of 150, 350, 450, 600 ml etc and thus simplify its production line.

Advantages:

- i) Simplification reduces variety, volume of remaining products may be increased.
- ii) Simplification provides quick delivery and better after sales service.
- iii) Simplification reduces inventory and thus results in better inventory control.
- iv) Simplification implies fewer parts and fewer the parts the lower the production costs.
- v) This reduces product cost.
- vi) Improves product quality.

Disadvantages:

- i) Not able to meet the need of wide range of customer's preferences.
- ii) Possibility of loosing customers because of competition.
- iii) Creates conflict of interest among marketing and production team.

Applications:

- 1) Automobile vehicles range.
- 2) Computer workstation parts and accessories, eg pen drive, memory card etc.

Specialization:

- This is the natural outcome of the application of standardization and simplification. Specialization meat concentrating efforts on a specified field of action or towards a specific attempt.
- A worker is said to be specialized in his work, when he acquire skills and proficiency in it by concentrating solely on particular work or job. A mechanic is specialized in only particular type of vehicle or make.
- A factory producing only gearboxes or bearings is specialist in production eg NRB Bearing of Bosch Dusel Pumps.

Advantages:

- Workers achieve a high state of skill and proficiency.
- They take less times to complete the activity in which they are specialized.

Disadvantages:

- Specialized labour and equipment are not flexible i.e. they can not be used for other purposes.
- Specialization may result in monotony.

Applications:

- Specialization is universal in application, it is a rule rather than exception in today's industry.
- Specialization can be applied to

i) Products

ii) Processes

iii) Individuals

iv) Companies

v) Jobs

vi) Equipment's

Product Development Versus Product Design

- Product development is a complete cycle from market research, product specification, concept design, costing scheduling, testing, manufacturing, logistics, customer feedback, improvement and all other aspects of getting complete product into the market.
- Complete product development process includes involvement of marketing, design, manufacturing, sales and after sales support team.
- Whereas product design is a complete design process that include product industrial design, user experience. 3D CAD Design, design calculations and simulation Responsibility of a good product design is to ensure product working as per customer requirement.
- Product development in most cases leads to final Design. Most people assume that when a designer designs a product on a proper of in CAD software, that's the final product. However, it is not. This is where product development plays a role. When you start prototyping of a product, what looks pretty in a sketch or computer rendering does not always functions in real environment.

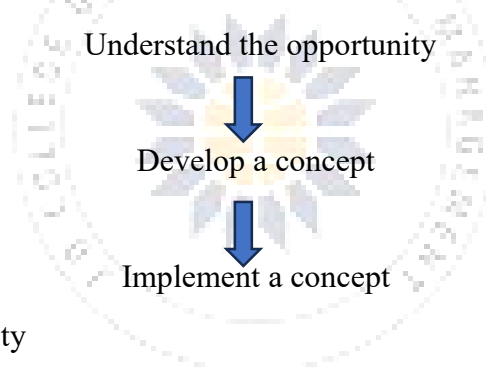
Product Development Includes	Product Design Includes
Market analysis	Synthesis (development of different alternatives)
Product specification	
Technologies research	Sketching
Products experience design	Analysis
Technical development	Selection-Selected the best one
Industrialization (Manufacturing)	Basic Engineering
Market launch	Apply Basic Rules
Analysis and improvement	Product Design Prototype for testing purpose Manufacturing

	Operation Collecting the data or feedback during the operation. Product development
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- Product development is the last step of product designs. After launching the product into the market, if we will see any factors by which we can enhance the quality of our product in terms of efficiency, cost reduction etc. Then we will modify our product and that's called product development process.
- So product development is enhancing or improving the functions of existing product.

Modern Product Development Process

Product development is a process of set of activities and tasks. These are tasks of creating tasks, tasks of understanding, task of communication, task of testing and task of persuasion. This mainly consists of three phases



i) Understand the opportunity

a) Develop a vision

b) Market opportunity analysis

c) Customer need analysis

d) Competitive analysis

a) Develop a vision

- This is the first step in product development for a new product. There are certain questions need to be answer to develop a vision for new product.
- The answer for above questions are vision for new product.

b) Market opportunity analysis The question is whether any vision can be transferred into successful realization. Can it be developed and implemented into a product at worthwhile profit? The answer for above question is market opportunity, the revenue that might be generated by a new product from a market has to be estimated. In todays competitive environment the estimated price and volume, the market will carry is always the starting point.

c) Customer need analysis: After completing market analysis and made a decision to proceed to develop new product a development team must analyze and understand what customer wants new product to do?

d) Competitive analysis: Once customer needs are understood, the competitive products on the market must be analyzed in terms of how well they satisfy the customer needs.

2) Develop a concept

- a) Portfolio planning b) Functional modeling
- c) Product architecture d) Concept engineering

a) Portfolio planning: After gathering and verifying all available information on what a new product must do, how it must fit into the market and what cost range constrains, the product, the design team can work to meet these expectations.

The first activity in this efforts is to design a set of general market specification for the product.

This activity must be undertaken with respect to the complementary set of products that the company has to offer.

b) Functional modeling After understanding portfolio architecture, customer needs and competition design team starts work on new concept generation. One of the first task in concept generation is to determine, what the product must do to supply the customer satisfaction.

c) Product architecture development: All functional model describing the inputs, output and transformations that must happen for a product to work, there are possible grouping of subsets of these functions into actual subassemblies.

d) Concept engineering: The functional model and alternative product architecture set the stage for very effective concept reengineering. At this stage product development team generates many concepts for implementing the functional specifications. After synthesizing many concepts, a design team must select one best to implement.

3) Importance of concept:

- a) Embodiment engineering b) Physical and analytical modeling
- c) Design for X d) Robust design

a) Embodiment engineering: After selecting a concept, it must be implemented, which is the final phase of product development. In this stage, a chosen concept is given forms through specifications of components to be purchases parts to manufacture and specifications for their assembly into the product.

b) Physical and analytical modeling: One important aspect of embodiment engineering is modeling and testing of new implementation ideas by physical construction.

- In product development, it does not require to explain physical phenomenon. The aim is to develop a product that will delight customer, fulfill the needs of design process.
- Therefore it is required to consider modelling in the real world context. This demonstrate how it support design decision making.
- The method to actually make model of the physical processes in product are difficult to initiate for new and experienced engineers.
- Thus, develops a methodology for construction a mathematical model of customer needs in terms of variables that a design team can change.

c) Design for "X": In addition to many performance test of a product, design team has additional engineering specification that their product must meet i.e. Design for "X", where 'X' is one of these requirement. E.g. Design for manufacturing and assembly, design for environment, design for robustness etc (refer unit 5 for detailed answer).

d) Robust design: The design team must place on making the product work well. Any untrained person can make something but it takes analysis. Every product of the assembly line works properly and whose each work well in varying applications and conditions. Ensuring this quality view of a product is the goal of robust design.

Product Testing and Validation

- **Product testing:** Manufacturer by own or any third party agency or institution. A product prototype is required to be developed and initial testing required to be performed on prototype. Manufacturer need to initiate trial or pilot production in order to test.
- In this process, it is required to assess whether the product finds satisfactory market acceptance. It is crucial to test whether the product and marketing concept satisfy the consumers in market place.
- Testing the product in real market environment allows designer or company to directly interact with customers, who have not been screened priorly. If customer is allowed to test the product in a real life situation, they will tell design team whether they are truly satisfied with product or not.
- Many times third party testing agencies have been asked to conduct confirmatory and regulatory test for product certification and licensing or permit.

Product Validation and Verification

- Verification is a paper exercise which starts with taking all the design inputs, specifications, government and industry's self regulations (Homologation).
- Knowledge taken from previous designs and any other information necessary for proper function are required to compare product design outputs, drawings, assembly instructions and digital drawing files.
- In comparison, it is ensured that each requirement in the input is accounted and verified in the design output.
- The output of this verification review is often recorded in a statement of compliance document. This document will list every requirement for the design, identify if the design is compliant or not.
- The obvious importance in this step is to make sure the design has not missed addressing any requirements. If the requirement is not compliant means design does not meet a requirement.

Design Validation:

- Validation is a step where design team actually build a version of product and validation be done against the requirements as modified after verification process.
- This does not necessarily mean the first production unit but it can also be an engineering model which some company used to prove the first run of complicated new design.
- This assures that a product meets the needs of the customers and other identified stakeholders.

Example for Verification

- The output of this verification review is often recorded in a Statement of Compliance document. This document will list every requirement for the design, identify if the design is compliant or not, and list where this compliance is proven in the documentation.

Requirement	C/NC	Verification
Max Dimensions 1" x 2" x 4"	C	Drawing 123456-7

#C-Compliance

#NC-Not Compliance

- The obvious importance in this step is to make sure that the design has not missed addressing any requirements. If requirements are not met, meaning the design does not meet a requirement, now is the time to know this and negotiate with the customer if this requirement is necessary or can be relaxed.

Example for Validation

- This will often require more testing that will be used on production models. To ensure that all requirements are met, a full set of measurement and tests is done on the validation unit.

Requirement	C/NC	Verification	Value	Validation
Max Dimensions 1" x 2" x 4"	C	Drawing 123456-7	0.95" x 1.98" x 3.85"	CMM Report Order 98765

#C-Compliance

#NC-Not Compliance

- After validation, the full set of requirements on one unit of most products can have a reduced level of inspection and testing, depending on factors such as requirements criticality or manufacturing process capability. A good product validation can help decide which requirements need to be checked on every product, and which do not.

Questions

- Define product design. Enlist and explain factors affecting product design.
- Discuss standardization, simplification and specialization in context to product design.
- Differentiate between product design and development.
- Explain various phases for modern product development process.
- Discuss product validation and verification.
- Explain factors to be considered for product design.
- Enlist and elaborate different factors affecting product design.
- How the product is supposed to do?
- What controls of the products will be automatic or manual?
- Explain the relationship of standardization and specialization with product design.
- What product do we wish were out there?
- What is the difficulty with the current product we use?
- Why does it not do something, we want it to do?
- Explain product testing.
- Explain product verification and validation in detail.
- Discuss product verification and validation with suitable examples.

17. Whether each required test called out in the test instructions including the correct pass/fail criteria for each test?
18. Whether all product acceptance criteria correct?
19. Whether all physical characteristics identified in the build instructions?

